

# Robust Medical Image Segmentation with Limited Annotations Problem

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**Abstract**— Medical image segmentation is a fundamental task in computer aided diagnosis and clinical analysis. However, developing accurate segmentation models remains challenging due to the limited availability of annotated medical images, as manual labeling requires significant time and expertise from medical professionals. Traditional supervised learning approaches rely heavily on large labeled datasets, which are often difficult to obtain in medical imaging applications. In this study, we investigate the effectiveness of semi supervised learning techniques for cardiac image segmentation. Our approach leverages both labeled and unlabeled data to improve segmentation performance. Several semi supervised methods, including Mean Teacher, Cross Pseudo Supervision (CPS), adversarial learning, and the diffusion-based rectification method DiffRect, are evaluated on the ACDC Dataset. We further compare these approaches with fully supervised training to assess their effectiveness in scenarios with limited labeled data.

**Keywords**—Medical imaging, Segmentation, Semi-supervised learning.

## I. INTRODUCTION

Medical image segmentation has become a fundamental task in computer vision, particularly in the field of computer-aided diagnosis and clinical decision support. The main objective of medical image segmentation is to accurately identify and delineate anatomical structures or regions of interest within medical images [1]. With the rapid advancement of deep learning techniques, supervised learning approaches have achieved remarkable success in medical image analysis tasks, providing high accuracy and efficient processing. Architectures such as U-Net [2] have become widely adopted due to their strong performance in biomedical segmentation problems.

Despite these advances, supervised learning methods heavily rely on large amounts of annotated data. In medical imaging, obtaining high-quality annotations is particularly challenging because it requires significant time and expertise from medical professionals [1]. Consequently, the scarcity of labeled medical datasets limits the performance and generalization ability of deep learning models. To address this limitation, semi-supervised learning has emerged as a promising approach that enables models to leverage both labeled and unlabeled data during training.

In this work, we focus on cardiac image segmentation using the ACDC Dataset. Several semi-supervised learning methods are investigated to improve segmentation performance under limited labeled data conditions. These

approaches include Mean Teacher [3], Cross Pseudo Supervision (CPS) [4], adversarial learning strategies inspired by Semi-Supervised Learning with Generative Adversarial Networks [5], and the diffusion-based rectification method DiffRect [6].

The main contributions of this work are threefold:

- We test and evaluate several semi-supervised segmentation methods for cardiac MRI analysis.
- We compare the performance of these approaches with a fully supervised baseline under limited labeled data conditions.
- We analyze the effectiveness of diffusion-based rectification techniques for improving segmentation accuracy.

The remainder of this paper is organized as follows. Section 2 reviews related work in medical image segmentation and semi-supervised learning. Section 3 describes the experimental setup and dataset description. Section 4 reports the quantitative and qualitative results and provides a discussion of the findings. Finally, Section 5 concludes the paper.

## II. RELATED WORK

### A. Medical image segmentation

Deep learning has significantly improved the performance of medical image segmentation systems. Architectures such as U-Net have become a standard approach due to their encoder-decoder structure, which enables the extraction of hierarchical features while preserving spatial information. A key component of U-Net is the skip connections [7] that link encoder and decoder layers. These connections allow low-level spatial details from early layers to be combined with high-level semantic information during reconstruction, thereby improving segmentation accuracy.

More recently, efficient backbone networks such as EfficientNet [8] have been integrated into segmentation frameworks to enhance feature extraction while maintaining computational efficiency. EfficientNet optimizes network performance through a compound scaling strategy, which balances network depth, width, and input resolution to achieve high accuracy with fewer parameters and lower computational cost.

By leveraging these optimized feature representations within a U-Net-style architecture, models such as EfficientUNet [9] achieve accurate segmentation while

maintaining fast inference and efficient use of computational resources. In this architecture, EfficientNet is used as the backbone of the encoder, replacing the traditional convolutional encoder of U-Net, while the decoder retains the characteristic skip connections to progressively recover spatial details and generate precise segmentation maps.

### B. Semi-supervised medical image segmentation

To address the scarcity of labeled medical data, semi-supervised learning approaches have been widely investigated. Among these approaches, consistency regularization methods are commonly used to enforce stable predictions under different perturbations of the input data.

For instance, the Mean Teacher [3] framework encourages consistency between a student network and a teacher network whose weights are updated using exponential moving averages.

Another widely used strategy is mutual learning, where multiple networks learn collaboratively by exchanging pseudo-labels during training. For example, Cross Pseudo Supervision (CPS) [4] trains two segmentation networks simultaneously and uses predictions from one network to supervise the other, improving generalization when labeled data is limited.

Adversarial learning has also been explored to leverage unlabeled data effectively. Inspired by generative adversarial learning frameworks such as Semi-Supervised Learning with Generative Adversarial Networks [5], these methods introduce a discriminator network that distinguishes between reliable and unreliable segmentation outputs, encouraging the segmentation network to produce more realistic predictions.

More recently, diffusion-based techniques have been introduced to further refine segmentation results. In particular, DiffRect [6] employs a diffusion process to progressively correct segmentation predictions, leading to improved segmentation quality.

### C. Benchmarking semi-supervised segmentation methods

Motivated by these advances, this work provides a comparative evaluation of several representative semi-supervised learning strategies for cardiac MRI segmentation.

Specifically, we benchmark consistency-based, mutual learning, adversarial, and diffusion-based approaches under limited labeled data conditions in order to analyze their effectiveness and limitations

## III. EXPERIMENTS

### A. Dataset description

The dataset used in this study is the Automated Cardiac Diagnosis Challenge (ACDC) dataset, which contains cardiac cine MRI scans of 100 patients acquired at end-diastole (ED) and end-systole (ES). Each scan includes manual annotations of three structures: the left ventricle (LV), right ventricle (RV), and myocardium (MYO). The dataset covers a variety of cardiac conditions, including normal and pathological cases, providing a challenging benchmark for segmentation algorithms.

The data split was performed at the patient level, with 80 patients used for training, 20 patients for validation, and an additional 50 patients for testing. The scans were preprocessed and stored as H5 files to facilitate efficient data loading during

training. To support 2D model training, slices were extracted from the 3D MRI volumes, resulting in four main data lists:

- **train\_slices:** 2D slices extracted from the training patients, used for supervised and semi-supervised learning.
- **train:** full 3D MRI volumes of the training patients.
- **val:** 3D MRI volumes reserved for validation, used to monitor model performance and prevent overfitting.
- **test:** 3D MRI volumes of the test patients, used for final evaluation.

The ground truth for all splits was obtained from the manual annotations of the MRI scans, which served as supervision for training and validation. An example of an MRI slice from ACDC dataset, its corresponding ground-truth mask, and the scribble annotation is shown in Fig. 1.

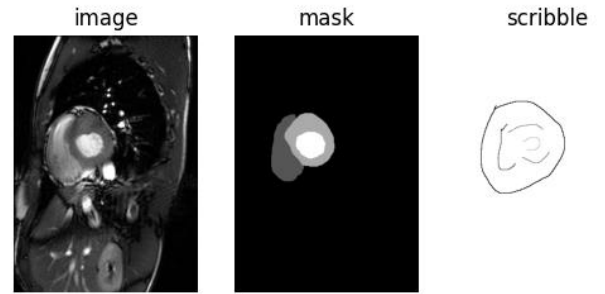


Fig. 1. Example slice from the ACDC dataset with its full segmentation mask and the corresponding scribble annotation illustrating weak supervision..

### B. Evaluation metrics

To analyze the segmentation performance objectively, three evaluation metrics were used: Dice Similarity Coefficient (DSC), 95th percentile Hausdorff Distance (HD95), and Average Surface Distance (ASD). These metrics evaluate both region overlap and boundary accuracy between the predicted segmentation and the ground truth.

The evaluation was performed on the three anatomical structures: left ventricle (LV), myocardium (MYO), and right ventricle (RV). The background class was not included in the metric computation. Metrics were first computed separately for each class and each test volume, then averaged across all test patients. Finally, the mean performance over the three anatomical structures was reported.

- **Dice Similarity Coefficient (DSC):**

The Dice coefficient measures the overlap between the predicted mask and the ground-truth mask. It is defined as:

$$Dice = \frac{2|P \cap G|}{|P| + |G|} \quad (1)$$

where:

- P represents the predicted segmentation,
- G represents the ground-truth segmentation.

- **Hausdorff Distance (HD95):**

The Hausdorff Distance measures the maximum distance between the boundary points of the predicted and ground-truth segmentations. However, to reduce sensitivity to outliers, the 95th percentile of the Hausdorff Distance (HD95) is used instead of the maximum value.

- **Average Surface Distance (ASD):**

The Average Surface Distance computes the average distance between the surface points of the predicted segmentation and the ground-truth segmentation.

### C. Data processing

All experiments were conducted using a unified preprocessing pipeline to ensure a fair comparison between the evaluated architectures. The dataset was stored in H5 format, where each file contains the MRI slice and its corresponding segmentation mask. Training was performed using 2D slices extracted from the 3D MRI volumes. During training, each slice was resized to a fixed spatial resolution of  $256 \times 256$  pixels using nearest-neighbor interpolation for both images and labels. This standardization ensures consistent input dimensions across all evaluated models. To improve model generalization and reduce overfitting, data augmentation was applied during training. The augmentation strategy included random 90-degree rotations, random horizontal and vertical flips, and random rotations within the range of  $-20^\circ$  to  $+20^\circ$ . These transformations were applied jointly to the image and its corresponding label to preserve spatial alignment. No additional intensity normalization was applied during training, as the images were already preprocessed prior to storage. The same preprocessing pipeline was used for all architectures to guarantee that performance differences arise solely from model design rather than data preparation.

DiffReCT extends this standard pipeline by incorporating a weak and strong augmentation strategy, which is particularly important for this architecture. For each training sample, a weakly augmented image is created using mild transformations, such as small rotations and flips, that maintain anatomical consistency. In parallel, a strongly augmented image is generated using more aggressive transformations, including brightness and contrast adjustments, sharpening, and optional Gaussian blur. This dual augmentation allows the network to learn consistency between predictions on weakly and strongly augmented inputs [6], effectively leveraging unlabeled data to improve segmentation performance.

### D. Implementation

- *Semi-supervised training*

All architectures were trained under a unified experimental setup to ensure fair comparison. Training was performed for a maximum of 10,000 iterations with a batch size of 8, of which 4 samples per batch were labeled. The backbone model for most experiments was an Efficient U-Net, chosen for its balance of representational capacity and computational efficiency. DiffReCT, however, uses a standard U-Net backbone, as its design is focused on leveraging weak and strong augmentations rather than model efficiency.

For each architecture, two separate models were trained to study the effect of labeled data quantity. One model was

trained with 7 patient labeled samples, representing a highly limited supervision scenario, while the other was trained with 14 patient labeled samples, providing moderately more supervision. In every architecture, including DiffReCT, two models were maintained: one for the low-labeled scenario and one for the higher-labeled scenario. This setup allowed a direct evaluation of how each method benefits from increasing labeled data.

In the case of DiffReCT, each batch used not only the labeled samples but also weakly and strongly augmented versions of the images. The weakly augmented images served as a stable reference, while the strongly augmented images were used to enforce consistency in the network's predictions in the latent deature rectification module.

The implementation of the semi-supervised segmentation methods used in this study is based on publicly available research code. In particular, the implementations of the Fully Supervised, Mean Teacher, Cross Pseudo Supervision (CPS), and Adversarial Network approaches were adapted from the official repository SSL4MIS: <https://github.com/HiLab-git/SSL4MIS/>. In addition, the implementation of the DiffRect method was based on the official repository DiffRect provided by the CUHK-AIM research group: <https://github.com/CUHK-AIM-Group/DiffRect/>

- *Supervised training*

For comparison, a fully supervised baseline was trained using the same preprocessing and training setup as the other methods. Three separate models were trained to evaluate the impact of labeled data quantity on performance. The first model was trained with 7 patient labeled samples, representing an extremely low-data regime. The second model used 14 patient labeled samples, matching the semi-supervised setup in the earlier experiments. Finally, a third model was trained with 80 patient labeled samples, representing the full labeled dataset scenario.

All baseline models used the Efficient U-Net backbone and were trained for a maximum of 10,000 iterations with a batch size of 8, identical to the semi-supervised experiments. The purpose of this baseline was to quantify the improvement gained from leveraging unlabeled data with semi-supervised methods.

## IV. RESULTS AND DISCUSSIONS

### A. Results

Experiments were conducted using two labeled data fractions, 7 and 14 labeled samples, to analyze the behavior of each method under limited annotation settings. The quantitative comparison between all evaluated methods, including Fully Supervised, Mean Teacher, CPS, Adversarial Network, and DiffRect, is summarized in [Table 1](#), which reports the Dice, HD95, and ASD values for each configuration, the gain achieved by each method is reported in [Table 2](#), qualitative comparison of segmentation results for four cardiac classes on three patients (frame 1). For the four Semi-supervised methods trained with 7 and 14 labeled patients, and the Efficient-UNet baseline is trained with 7, 14, and 80 patients are reported in [Fig. 2](#).

TABLE I. QUANTITATIVE EVALUATION OF DIFFERENT OBJECT SEGMENTATION METHODS

| Method       | Scan Used |           | Metrics |       |      |
|--------------|-----------|-----------|---------|-------|------|
|              | Labeled   | Unlabeled | Dice    | H95   | ASD  |
| Supervised   | 7         | 0         | 78.5    | 10.26 | 3.83 |
|              | 14        | 0         | 81.35   | 4.84  | 1.57 |
|              | 80        | 0         | 89.62   | 1.98  | 0.62 |
| Mean Teacher |           |           | 79.02   | 12.92 | 3.99 |
| CPS          | 7         | 73        | 78.97   | 11.60 | 3.76 |
| GAN          |           |           | 76.31   | 14.63 | 4.65 |
| DiffRect     |           |           | 81.91   | 5.70  | 1.53 |
| Mean Teacher | 14        | 66        | 81.51   | 6.53  | 2.16 |
| CPS          |           |           | 81.97   | 7.65  | 2.26 |
| GAN          |           |           | 77.97   | 10.28 | 3.15 |
| DiffRect     |           |           | 86.22   | 4.25  | 1.18 |

The experimental results demonstrate a clear and consistent superiority of DiffRect over competing semi-supervised learning strategies in low-label cardiac segmentation. The supervised baseline behaves as expected, with performance steadily increasing as the number of labeled samples grows (78.5 Dice with 7 labels, 81.35 with 14 labels, and 89.62 with full supervision), confirming the reliability and stability of the training framework.

In contrast, traditional consistency-based methods such as Mean Teacher and CPS yield only marginal improvements over the supervised baseline in both low-label settings. Their gains remain below one Dice point and are often accompanied by degraded boundary metrics (H95 and ASD), suggesting that the unlabeled data is not effectively leveraged to refine structural predictions.

The adversarial approach exhibits clear instability, with consistent performance degradation across both Dice and boundary metrics. This behavior indicates optimization difficulties and highlights the sensitivity of adversarial training in small-data regimes.

TABLE II. GAIN OF DIFFERENT OBJECT SEGMENTATION METHODS

| Method       | Scan Used |           | Metrics gain |       |       |
|--------------|-----------|-----------|--------------|-------|-------|
|              | Labeled   | Unlabeled | Dice         | H95   | ASD   |
| Mean Teacher |           |           | +0.52        | +2.66 | +0.16 |
| CPS          | 7         | 73        | +0.47        | +1.34 | -0.07 |
| GAN          |           |           | -2.19        | +4.37 | +0.82 |
| DiffRect     |           |           | +3.41        | -4.56 | -2.30 |
| Mean Teacher | 14        | 66        | +0.16        | +1.69 | +0.59 |
| CPS          |           |           | +0.62        | +2.81 | +0.69 |
| GAN          |           |           | -3.38        | +5.44 | +1.58 |
| DiffRect     |           |           | +4.87        | -0.59 | -0.39 |

DiffRect, however, achieves substantial and consistent improvements across all evaluation metrics. With 7 labeled samples, it provides a +3.41 Dice gain and significantly reduces boundary errors (-4.56 H95 and -2.30 ASD). With 14 labeled samples, the improvement further increases to +4.87 Dice, while maintaining superior boundary precision. Importantly, the gains scale coherently with the number of labeled samples, suggesting that DiffRect effectively integrates unlabeled information to enhance structural consistency and segmentation smoothness.

## B. Discussion

These observations can be partially explained by the learning mechanisms underlying each method. Consistency-based approaches such as Mean Teacher and CPS rely heavily on the quality of pseudo-labels or teacher predictions generated during training. When the number of labeled samples is extremely limited, the initial model may produce uncertain or noisy predictions, which restricts the effectiveness of the consistency constraint and limits the ability of these methods to fully exploit the unlabeled data. This issue is often referred to as reliance risk, where the model progressively reinforces inaccurate pseudo-labels. In cardiac segmentation, this problem can be particularly critical because the anatomical structures—such as the left ventricle (LV), right ventricle (RV), and myocardium—are strongly correlated and spatially dependent. Errors in the prediction of one structure may therefore propagate to the others during training.

Similarly, adversarial learning frameworks are known to be sensitive to optimization dynamics and require careful balancing between competing objectives, which may lead to unstable behavior in small-data regimes.

In contrast, DiffRect introduces a rectification mechanism that helps mitigate the impact of unreliable pseudo-labels, allowing the model to learn more robust structural representations from unlabeled samples. In particular, the Label Context Correction (LCC) block refines pseudo-labels before they are used as supervision by incorporating semantic and spatial context. This mechanism leverages anatomical relationships between structures—for example, the spatial organization between the LV, RV, myocardium, and the surrounding background—to correct inconsistent predictions and produce more coherent segmentation maps.

Additionally, DiffRect addresses the problem of distribution misalignment between labeled and unlabeled data. While traditional methods primarily focus on aligning predictions with ground-truth labels, the Latent Feature Rectification (LFR) module aims to align the distribution of latent feature representations, rather than relying solely on the distribution of ground-truth annotations. By encouraging consistency in the latent feature space, LFR helps the model learn more stable and generalizable representations, which improves segmentation performance in limited-label scenarios.

It should also be noted that the experiments were conducted under constrained computational resources, using a fixed backbone architecture and a limited number of training iterations. Under different experimental conditions—such as such as stronger backbone networks, extended training schedules, or further hyperparameter tuning—some of the evaluated semi-supervised methods could potentially achieve improved performance. Nevertheless, within the current experimental setup, DiffRect demonstrates the most reliable capacity to leverage unlabeled data and improve segmentation quality in low-annotation scenarios.

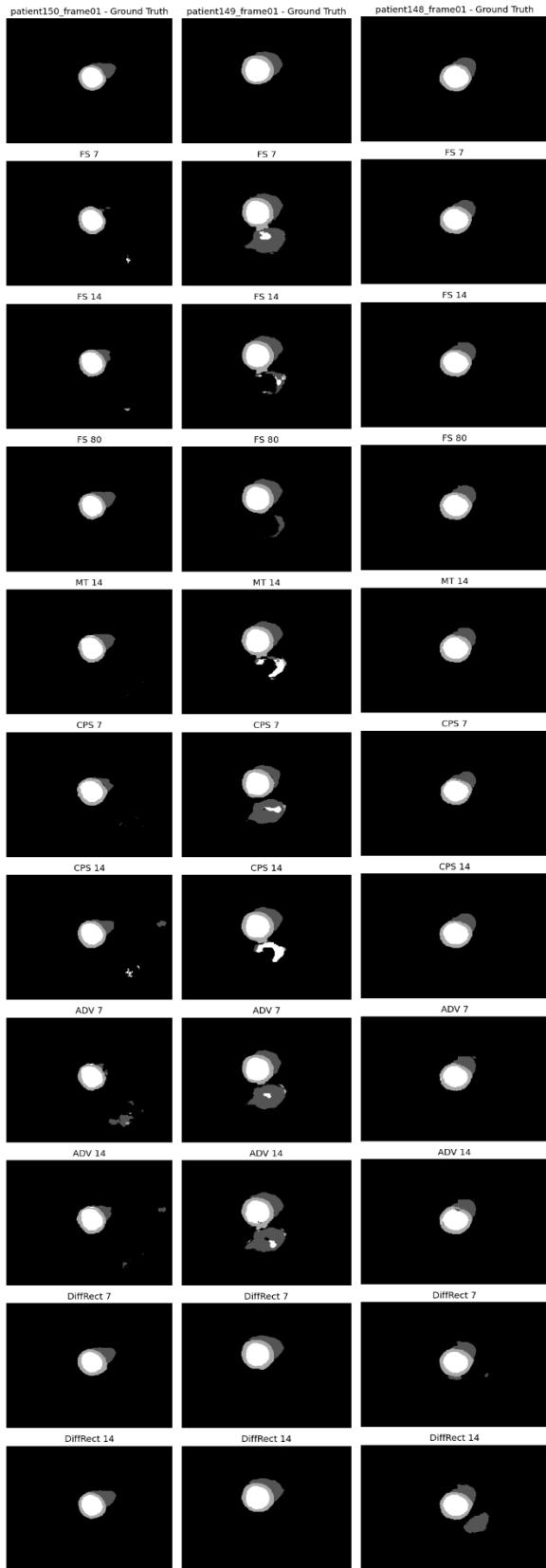


Fig. 2. Qualitative comparison of segmentation results for four cardiac classes on three patients (frame 1). Semi-supervised methods are trained with 7 and 14 labeled patients, while the Efficient-UNet baseline is trained with 7, 14, and 80 patients.

#### CONCLUSION AND PERSPECTIVES

In this work, we investigated the application of semi-supervised learning techniques for medical image

segmentation, focusing on improving performance when labeled data is limited. Several approaches were studied and experimentally evaluated on the ACDC dataset, including fully supervised training and different semi-supervised strategies such as Mean Teacher, Cross Pseudo Supervision, adversarial learning, and the diffusion-based method DiffRect.

The experimental results demonstrated that semi-supervised methods can effectively leverage unlabeled data to improve segmentation accuracy and robustness in low-annotation scenarios. Overall, this study highlights the potential of semi-supervised learning to address the scarcity of annotated medical data and to support the development of more efficient and reliable medical image analysis systems.

As future work, further investigations could explore the integration of stronger backbone architectures and longer training schedules to enhance model performance. Additionally, evaluating these methods on larger and more diverse medical imaging datasets, as well as exploring hybrid semi-supervised strategies that combine multiple learning paradigms, could provide deeper insights into their generalization capabilities in real clinical settings.

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